AegisLife Insurance Pvt. Ltd.

Risk Analytics and Claims Intelligence – Executive Summary Report

1. Project Overview

This capstone project was conducted for AegisLife Insurance Pvt. Ltd. to analyze claims risk, fraud patterns, policy lapse trends, and regional performance using SQL, Python, Excel, Statistics, and Power BI.
The objective was to build an integrated analytics framework to support executive decision-making and operational monitoring

2. Business Problem Statement

- High claim ratios impacting profitability
- Potential fraudulent / anomalous claims
- Lack of real-time decision dashboards

3. Key Performance Indicators

Total Claims	Avg Claim Amount	Fraud Ratio	Approval Rate
1406	252.93K	49.00%	35.85%

Business Interpretation

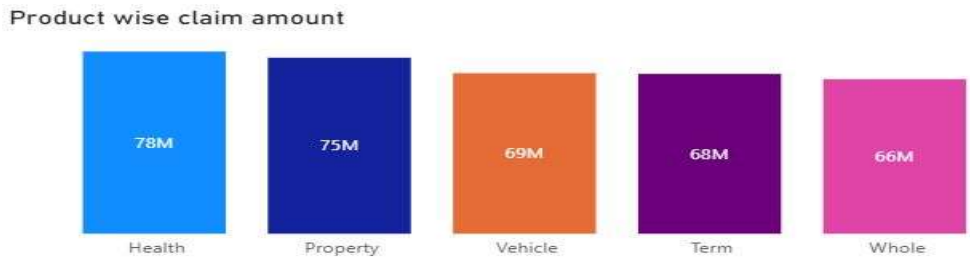
- Overall claims processed across all product lines
- Indicates moderate-to-high claim severity
- High fraud exposure; urgent anomaly investigation needed
- Low approval ratio; process review required

4. Region-wise Claims Distribution

Region	Total Claims	Share (%)
West	261	18.6%
Central	254	18.1%
East	237	16.9%
South	232	16.5%
North	230	16.4%
North-East	192	13.7%

Insight:
West and Central regions contribute the highest claim volumes, indicating higher operational and risk-monitoring pressure.

5. Product-wise Claim Exposure



6. Agent Performance Snapshot

Top 5 Best Agents

Agent ID	Retention Rate
AGT5002	100%
AGT5094	100%
AGT5103	100%
AGT5137	100%
AGT5162	100%

Bottom 5 Agents

Agent ID	Retention Rate
AGT5190	70.59%
AGT5154	70.65%
AGT5168	70.69%
AGT5264	70.74%
AGT5079	70.93%

7. Key Findings Summary

Area	Key Finding
Fraud	49% fraud ratio indicates urgent review
Retention	32.32% lapse rate suggests customer churn risk
Region	West & Central highest claim concentration
Product	Health line highest claim severity

8. Strategic Recommendations

1. Introduce region-specific fraud audit controls in West and Central regions to address the high fraud ratio (49%) and elevated claim concentration.

2. Reassess pricing, claim approval rules, and underwriting policies for Health insurance products, as this segment shows the highest total claim exposure (₹77.78 million).

3. Launch targeted agent retraining and customer retention improvement programs for low-performing agent portfolios, particularly those with retention rates below 75%, to reduce the overall policy lapse rate (14.90%).

9. Estimated Business Impact

• 8–12% projected reduction in claim leakage through improved fraud detection and regional audit controls

• 5% projected improvement in customer retention through agent performance enhancement and lapse reduction initiatives

• 2–3 days projected reduction in average claim processing time through process optimization and operational monitoring

10. Conclusion

The analysis identifies **fraud exposure, lapse risk, and product-level claim severity** as the primary business focus areas, enabling data-driven decisions for underwriting, retention, and operational governance.